

Integrating Waterfall and Agile Methodologies in Data Analysis: A Case Study of Credit Score Classification

INTRODUCTION

Credit scoring is one of the most influential tools in modern finance that has a major say on whether a person gets a house, a job, or a loan with good terms. However, even after having such a great impact on the financial futures of individuals, credit scores are still a mystery to a large portion of the public. Traditional research mainly focuses on easily measurable financial metrics like income and outstanding debt, but new studies reveal that behavioral and demographic factors such as payment delay patterns, age demographics, credit utilization behavior, and inquiry frequency have a significant impact on credit score classification (Hand & Henley, 1997; Grunert et al., 2005).

Handling a project that delves into such complicated issues requires not just the knowledge of the subject matter but also a well-thought-out approach in managing the project.

Key Project Management Requirements:

- Define work clearly to ensure reproducibility and stakeholder alignment
- Adapt to emerging data insights that reveal unexpected patterns
- Deliver iterative value rather than waiting for final comprehensive analysis
- Maintain comprehensive documentation for future reference
- Engage stakeholders continuously throughout the project lifecycle

These competing demands create a challenging environment where neither traditional project management approaches nor purely agile methodologies prove sufficient on their own.

Problem Statement and Relevance

Most firms see project management as a binary decision to either accept the organization and the predictability of the Waterfall methodology or to go all the way with the flexibility and the responsiveness of the Agile frameworks (Serrador & Pinto, 2015). This false dichotomy proves particularly problematic for data analysis projects, which inherently involve both:

- **Need for architectural clarity** — understanding data schemas, analytical methods, success metrics.
- **Requirement for adaptive problem-solving** — responding to unexpected data patterns, emerging insights.

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Data visualizations projects usually start with quite clear and detailed technical requirements (exact datasets, computational tools, analytical methods), however, they also have exploratory phases in which surprising patterns arise and thus the teams change their strategy abruptly. The two modes of sequential planning and pure iteration do not solve this problem of dual nature efficiently.

The credit score analysis project was an example of such challenges. The project analyzed 100,000 observations across 28 variables to investigate how behavioral, financial, and demographic factors influence credit score classifications. The analysis employed:

- Exploratory data analysis (distribution analysis, correlation matrices)
- Statistical hypothesis testing (Kruskal-Wallis, Chi-Square, ANOVA)
- Machine learning classification (Random Forest)
- Interactive visualization (Tableau dashboards)

During project execution, the team encountered scope expansion, violations of statistical assumptions, tradeoffs in model interpretability, and ongoing challenges in balancing the delivery of insights with comprehensive documentation.

Thesis Statement

The findings indicate that effective project management in data analysis initiatives necessitates a deliberate integration of the structural discipline of Waterfall methodology with the adaptive flexibility of Agile, rather than exclusive reliance on either approach (Conforto & Amaral, 2016). Specifically:

- **Waterfall offers the necessary architectural scaffolding** which makes the requirements clear and saves the effort from being wasted on the work that is not aligned.
- **Agile approaches embedded within Waterfall phases** can enable rapid iteration, continuous incorporation of feedback, and adaptive problem-solving without losing strategic direction.
- **The integration of both methodologies at two different levels generates superior results** across project predictability, stakeholder satisfaction, data quality, and rigor of analysis.

LITERATURE REVIEW

It is vital to understand the appropriate application of both Waterfall and Agile approaches by considering their historical development, respective strengths and limitations, and any emerging evidence on hybrid approaches. The development of project management frameworks has reflected wider organizational challenges in managing uncertainty, complexity, and stakeholder expectation.

The Waterfall model was first debated by Winston Royce in his seminal 1970 paper whereby he was addressing the development of large software systems (Royce, 1970). Royce presented a sequential, phase-gate approach which was the general way complex projects were handled during the 70s, 80s, and 90s. The method broke the projects down into different stages:

- **Requirements** — Define what will be built
- **Design** — Plan how to build it
- **Implementation** — Construct the solution

- **Testing** — Verify functionality
- **Deployment** — Release to production

Each phase generated official deliverables that could be used as an input to the following phase. Such a method of work was especially effective for those projects in which requirements were clear, and changes were expensive, e.g., projects in the fields of aerospace, defense, and sizable infrastructure.

Why Waterfall Dominated:

The appeal of Waterfall lay in its predictability. Organizations could:

- Estimate project duration with confidence
- Allocate resources systematically
- Plan organizational changes around known delivery dates
- Establish formal acceptance criteria at each phase
- Prevent progression until prior work met quality standards.

This discipline was useful in situations where errors were costly and it was not possible to make changes at a late stage (Kerzner, 2017).

Nevertheless, with the rapid pace of software development in the 1990s and early 2000s, the shortcomings of Waterfall were becoming increasingly obvious. Software projects frequently encountered:

- Unanticipated complexity
- Shifting stakeholder priorities
- Emergent requirements only clear during implementation

These limitations being recognized led to the 2001 Agile Manifesto, which was written by Beck, Fowler, Highsmith, and other members (Beck et al., 2001). This manifesto fundamentally reoriented project management thinking by prioritizing:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

After the Agile Manifesto was published, frameworks such as Scrum and Kanban have been widely adopted by technology and business organizations (Schwaber & Sutherland, 2020). However, it is noteworthy that some companies still use Waterfall for their projects, and the latest studies show that hybrid approaches, which mix elements of both methodologies, have become the most common way of working and are often better than just choosing one method (Cohn, 2010).

Waterfall Methodology: Strengths and Limitations

Waterfall methodology provides substantial benefits when projects have well-defined requirements and predictable execution paths.

Key Strengths:

- **Clarity and Predictability** — Defining the scope, schedule, and resources to be used very clearly to the stakeholders gives them a clear idea of what to expect and reduces the number of surprises that they might get.
- **Documentation Discipline** — Detailed documentation makes it possible to look it up later, do the same thing, and transfer the knowledge. When team members leave, the most valuable knowledge is retained.
- **Structured Quality Gates** — Reviewing the different stages and having formal acceptance criteria help in identifying the errors that have been made early and in keeping the standards.
- **Resource Efficiency** — The defining of the different stages in a clear way makes it possible to allocate and release resources in a predictable manner.

Significant Limitations:

- **Inflexibility** - When a phase is finished, it is very expensive and unsettling to return to the previous phases, thus, there is a resistance to adaptive planning.
- **Delayed Feedback** - Stakeholders and end-users, in most cases, get to see the working products only at the end of the project, hence, their input is very limited until the late stages when making changes is very costly.
- **Risk Concentration** - In case that requirements are wrongly understood or changed during the execution, the full effect of this scenario will only become visible at the end of the project, thus it might be necessary to do costly rework.
- **Knowledge Bottlenecks** - If there is too much dependence on upfront requirements elicitation, it is very likely that some critical aspects that are only revealed during the analysis and implementation will be overlooked.

Agile Methodologies: Strengths and Limitations

Agile methods especially Scrum, split work into short cycles that last two to four weeks. Every cycle contains four fixed meetings - a daily stand up, a planning session, a review of the finished work and a retrospective (Highsmith, 2009).

Key Strengths:

- **Adaptive Learning** — Short cycles give teams the chance to identify new requirements, data issues, and analytical opportunities by adjusting the way they work quickly
- **Continuous Stakeholder Engagement** — Frequent sprint reviews and demos give stakeholders the possibility to see and influence the work continuously, thus improving their alignment and satisfaction
- **Incremental Value Delivery** — Intermediate insights, visualizations, or analyses of even incomplete projects deliver utility and generate feedback
- **Risk Mitigation** — The short planning horizons indicate that misunderstandings are visible within weeks instead of months and thus, the risk of compound problems is reduced.

Significant Challenges:

- **Scope Uncertainty** — The lack of clear definition upfront, projects might broaden endlessly or become that the project loses its strategic focus.
- **Documentation Debt** — The expression "working software over comprehensive documentation" may result in organizations garnering insufficient documentation for long-term maintenance.
- **Coordination Complexity** — Teams that are distributed or large may have difficulties with rapid changes in direction and priority.
- **Resource Variability** — Ongoing adaptation can confuse resource planning and capacity management to be harder than conventional methods (Martinsuo & Killen, 2014).

Hybrid Approaches and Research Evidence

Recent research papers acknowledge more and more that in practice, projects are hardly ever either completely Waterfall or completely Agile. Hybrid models are suggested by such frameworks as SAFe (Scaled Agile Framework) and DAD (Disciplined Agile Delivery), which keep Agile principles in the frameworks of structured governance.

Critical Research Finding:

Research conducted by Serrador and Pinto (2015) revealed that hybrid approaches mixing the planning discipline with the iterative delivery were more successful in most cases than either pure Waterfall or pure Agile implementations when measured by various success metrics:

- Schedule adherence
- Budget management
- Stakeholder satisfaction
- Project quality
- Team morale

Hybrid approaches are especially relevant for data science and analytics projects. Such projects often start with very clear technical requirements (specific datasets, computational tools, analytical methods) which are best handled by Waterfall due to its architectural clarity. However, these projects also have an exploratory phase in which the adaptability of Agile allows for a quick response to an unexpected data pattern or methodological insight. It is essential to realize this dual nature and create project management methods that use both methodologies for the best results.

CREDIT SCORE ANALYSIS PROJECT

OVERVIEW

The case study behind this research was about a public credit score dataset that had 100,000 observations and 28 variables. The project looked at the interrelations of behavioral, financial, and demographic factors with the credit score categories (Poor, Standard, Good) by means of the data set. Various methods such as exploratory data analysis, statistical hypothesis testing, machine learning classification, and interactive visualization were used in the analysis.

Dataset Characteristics and Research Questions

The dataset contained comprehensive customer financial and behavioral data:

Financial Variables:

- Annual income (highly skewed distribution requiring log transformation)
- Number of loans, bank accounts, credit cards
- Outstanding debt and credit limits
- Credit utilization percentages

Behavioral Variables:

- Payment history metrics (days delayed from due date)
- Credit inquiry frequency
- Credit limit changes
- Interest rates

Demographic Variables:

- Age (continuous, transformed into five life-stage categories)
- Occupation categories
- Employment information

The research team wanted to know the main factors that influenced the classification of credit scores. Such an understanding is very important for making lending decisions and improving consumer financial literacy (Arya et al., 2013).

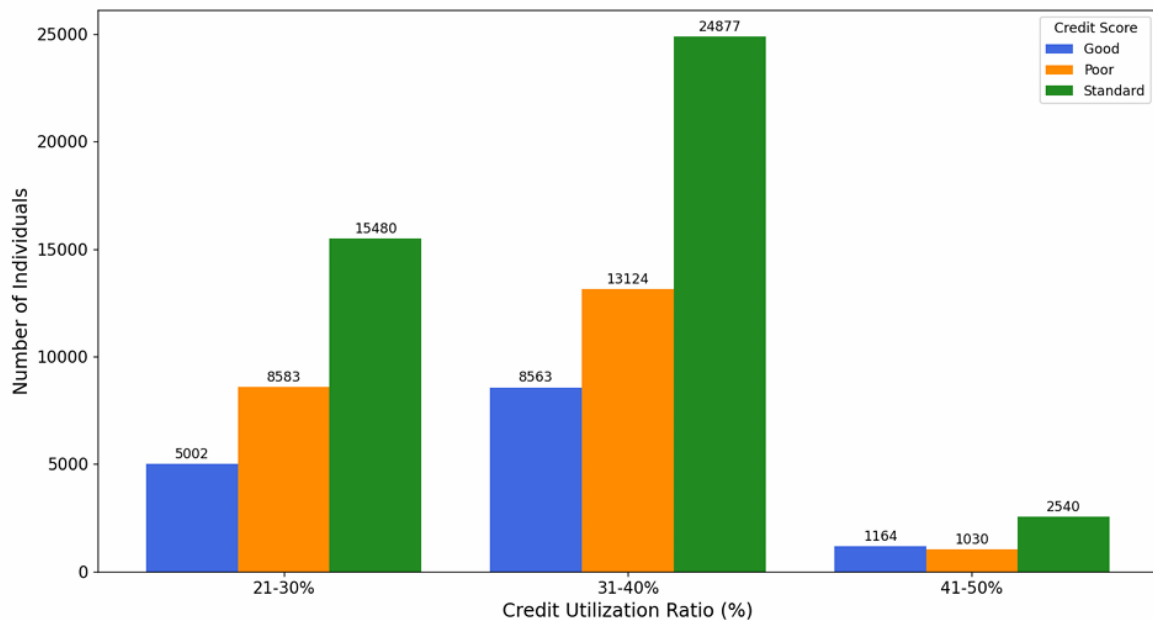
Five Core Research Questions:

1. Does annual income meaningfully correlate with credit score, and how much variance does it explain?
2. How significantly do payment delays impact score classification—do borrowers with longer delays consistently receive lower scores?
3. What role do age and life stage demographics play in credit scoring?
4. How influential is credit utilization behavior, and what utilization threshold predicts lower scores?
5. What is the relationship between inquiry frequency and score classification—does frequent credit seeking indicate financial distress?

Analytical Approach and Key Results

To find answers to these questions the group performed methodical exploratory data analysis to figure out the principal patterns, planned suitable statistical tests and machine learning models, and developed interactive visualizations that allowed the stakeholders to explore the results.

Statistical Test Results:



The **Random Forest Classifier** was able to correctly predict instances **72.8%** of the time, showing the most outstanding performance when predicting standard credit scores and less accurate predictions for extreme categories (Poor and Good). The result implies that credit scoring is the most challenging when the distribution is at the extremes.

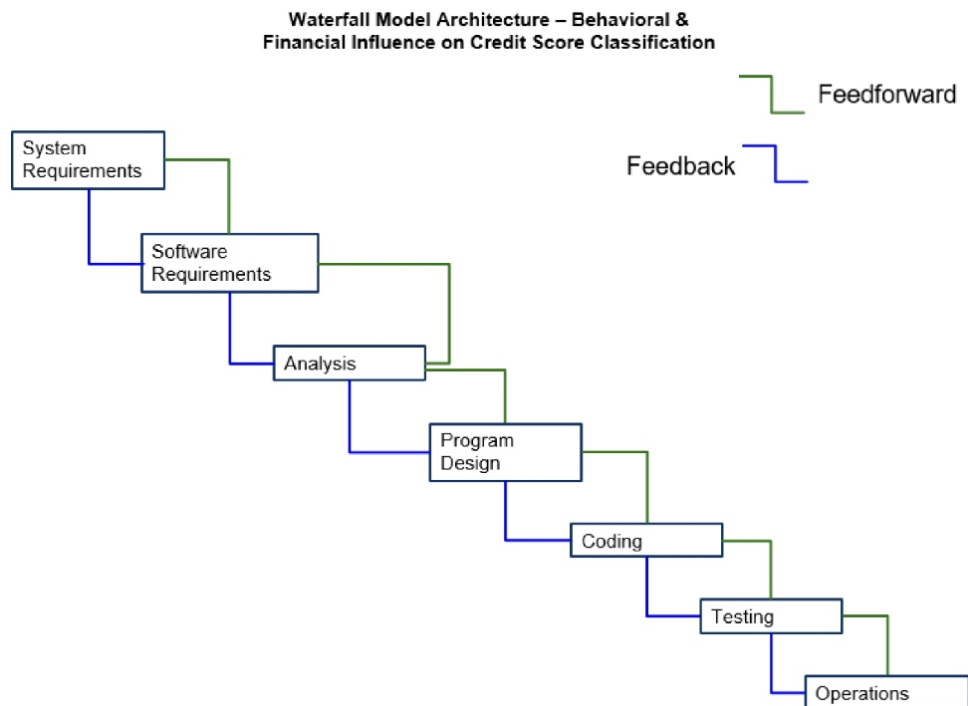
Key Findings:

- **Payment Delays** — Very significant differences were observed between credit score groups ($H(2)=15259.73$, $p<0.001$). Median delays: Good=10 days, Standard=17 days, Poor=26 days.
- **Age Demographics** — Strong associations with credit scores were found ($\chi^2(8)=362.12$, $p<0.001$). Middle-aged people (35-59) were the ones to have a significantly higher number of Good credit scores.
- **Credit Utilization** — Average utilization ratios changed significantly ($F(2,80362)=80.32$, $p<0.001$). Best utilization range: 21-30%.
- **Inquiry Frequency** — Credit scores were moderately negatively correlated ($r=-0.41$, $p<0.001$). Good scores were accompanied by a low number of inquiries; Standard/Poor categories by a high frequency.

These findings demonstrate that behavioral and demographic factors not just financial metrics but meaningfully predict credit scores (Grunert et al., 2005).

WATERFALL FRAMEWORK APPLICATION

The credit score analysis project utilized a seven-phase Waterfall layout that outlined different layers of the architecture, kept the scope from drifting, and marked out the clear deliverables for each stage. Although this might look as if it was too official for a single-team research project, the organization was of great help in controlling the scope, maintaining the quality, and making the sharing of knowledge easier.



Phase 1: System Requirements Definition

The very first stage was all about developing the main research goals of the study, figuring out the most important questions, and deciding how to measure the success.

Activities:

- Convened stakeholders (course instructor, peer reviewers, visualization specialists)
- Clarified research scope and aligned expectations
- Documented five core research questions
- Identified primary stakeholder groups
- Established deliverable expectations and success criteria

Outcomes:

- Project charter documenting research questions
- Stakeholder alignment on project direction
- Clear definition of what constitutes success

This fundamental work avoided confusions that arose later. When the team documented that the project would deal with only five questions, they had the possibility to look back at this charter each time new analyses were suggested: "This analysis on employment sector effects is not one of our core questions; do we take it up or leave it?" Clearly written records made these talks possible (Kerzner, 2017).

Phase 2: Software and Project Requirements

The second stage was about the in-depth data investigation and defining the technical specifications. The initial evaluation of data quality has unveiled few issues that have to be solved before further analysis.

Data Quality Issues Identified:

- **Income Distribution Skewness** — Annual income exhibited highly right-skewed distribution (skewness=26.252) with extreme outliers. Required log transformation, successfully reducing skewness to 0.132 and satisfying normality assumptions (Shapiro-Wilk test $p > 0.05$).
- **Missing Values** — Existed in some occupation categories, requiring decision-making about imputation vs. exclusion.
- **Outliers** — Outstanding debt and credit limits required identification and handling strategies.

Why Early Detection Mattered:

Early detection of data quality issues during Phase 2 prevented them from undermining subsequent analyses. Had the team discovered income distribution problems only after building regression models in Phase 5, correcting it would have required:

- Reverting to Phase 2
- Re-transforming data
- Rebuilding all subsequent analyses
- Substantially extending project timeline

This early detection prevented expensive rework downstream (Turner, 2009).

Phase 3: Analysis Planning

Phase 3 involved designing the analytical approach before implementing analysis, ensuring the team selected appropriate methods before consuming resources.

Statistical Method Selection:

For **payment delays analysis** — Kruskal-Wallis H test chosen because:

- Preliminary normality testing (Shapiro-Wilk) indicated violations of parametric ANOVA assumptions.
- Nonparametric Kruskal-Wallis test makes no distributional assumptions
- Still powerful for testing whether groups differ

For **age-credit score relationships** — Chi-Square test of independence chosen because:

- Both variables categorical
- Contingency table approach directly addresses association
- No assumptions violated

Documentation Discipline:

- All assumptions for each test documented
- Mitigation strategies for assumption violations identified in advance
- This documentation proved invaluable when violations were later discovered

Phase 4: Program Design

Program design required the specification of the data transformation, analytical, and visualization pipelines.

Key Design Decisions:

- **Data Transformation:** Natural logarithm for income (selected over alternatives like square root, Box-Cox)
- **Feature Engineering:** Five life-stage categories from continuous age
 - Young Adults 18-24
 - Early Career 25-34
 - Established Adults 35-44
 - Midlife Stability 45-59
 - Senior Citizens 60+
- **Machine Learning:** Random Forest configuration (100 trees, max depth 15, random state 42 for reproducibility)
- **Visualizations:** Specified formats before implementation
 - Bar charts comparing credit score distributions
 - Boxplots showing payment delay distributions
 - Feature importance plots showing top 10 contributing variables

The team through explicit visualization design in Phase 4 guaranteed the uniformity of visualizations and thus saved the implementation stage from making spontaneous decisions (Schwaber & Sutherland, 2020).

Phase 5: Coding and Implementation

Phase 5 executed the analytical procedures specified in Phase 4.

Implementation Activities:

- **Data Cleaning Pipelines** — Python using pandas and numpy libraries
- **Statistical Tests** — scipy and statsmodels libraries, with results captured in structured formats
- **Machine Learning** — scikit-learn for Random Forest with train-test split ensuring unbiased assessment
- **Visualizations** — Tableau and matplotlib, following Phase 4 specifications
- **Version Control** — Git enabled tracking of analytical decisions with descriptive commit messages
- **Quality Assurance** — Peer review of all code before commit ensured quality

Phase 6: Testing and Verification

Phase 6 systematically validated analytical outputs and assessed model performance.

Validation Procedures:

- **Model Performance** — Random Forest achieved 72.8% accuracy compared to:
 - Baseline random guessing: 33.3%
 - Simple rule-based approach: 46% (most common class)
- **Assumption Verification** — Explicit testing of statistical assumptions:
 - Shapiro-Wilk normality tests for continuous variables
 - Levene's test for homogeneity of variances
 - Chi-Square analysis verification of expected cell frequencies >5
- **Sensitivity Analysis** — Examined whether results held when using alternative statistical methods
- **Visual Spot-Checks** — Confirmed that plots accurately represented underlying data distributions

When assumption violations were discovered, the team documented them and justified their analytical choices transparently (Snowden & Boone, 2007).

Phase 7: Operations and Maintenance

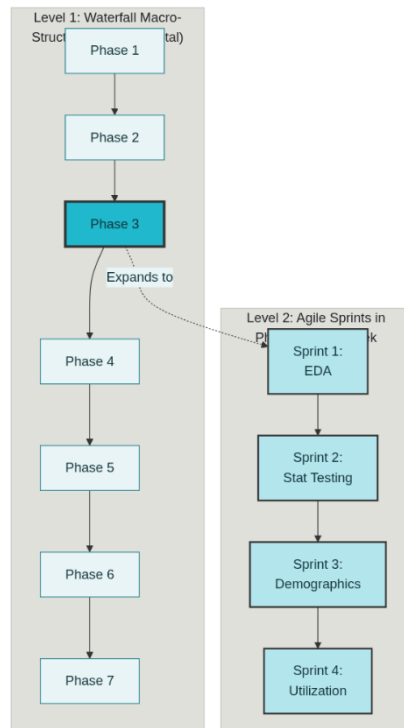
The final phase documented findings, reflected on limitations, and prepared materials for knowledge transfer.

Comprehensive Documentation Included:

- Explanations of every analytical decision
- Interpretation of results in business context
- Clear identification of limitations:
 - Random Forest's black-box nature prevented understanding individual feature interactions
 - Statistical test assumption violations (mitigated by large sample size $n=100,000$)
 - Missing variables (debt-to-income ratio, employment stability, income trends)
- Future research directions for extension

This documentation enabled knowledge transfer and prevented loss of critical information (Weick et al., 2005).

AGILE FRAMEWORK EMBEDDED WITHIN WATERFALL PHASES



While the total project organization was aligned with Waterfall's phase-gate discipline, Agile principles were incorporated into each phase through sprint-based cycles. The duration of each sprint was generally two weeks during which the team would focus on a single research question or an analytical component and produce a deliverable that would be independently valuable. This hybrid model made it possible to enjoy the benefits of Agile's adaptability and, at the same time, keep the structural discipline of Waterfall.

Sprint Structure and Minimum Units of Work

The concept of the "minimum units of work" (MUW) was fundamental in the functionalization of Agile in the project hierarchy. An MUW was the least unit which was independently valuable, and testable, and it made a direct contribution to the research question.

Example MUW #1 — Payment Delay Analysis Sprint

- **Owner:** Data Analyst
- **Duration:** 2-3 working days
- **Objective:** Determine if payment delays significantly impact credit score classification.
- **Tasks:**
 - Conduct exploratory analysis examining payment delay distributions across score categories
 - Execute Kruskal-Wallis H test comparing group medians
 - Interpret statistical results and conduct post-hoc Dunn's test
 - Create boxplot visualizations
- **Acceptance Criteria:**
 - Kruskal-Wallis test correctly executed with proper statistical reporting
 - P-values and effect sizes reported
 - Post-hoc pairwise comparisons documented

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- Visualization clearly represents underlying data distributions
- **Value Delivered:** Clear evidence about payment delay's importance, enabling stakeholder decisions about analysis direction.

Example MUW #2 — Age Demographics Analysis Sprint

- **Objective:** Understand how age influences credit scores and life-stage patterns
- **Tasks:**
 - Group ages into five life-stage categories based on financial development patterns
 - Produce cross-tabulations of age categories and credit scores
 - Execute Chi-Square test of independence
 - Create visualizations showing distribution patterns
- **Deliverables:** Cross-tabulation tables, statistical test results, visualizations showing proportions of Good/Standard/Poor scores in each age group

Example MUW #3 — Credit Utilization Analysis Sprint

- **Objective:** Examine whether utilization ratios differ significantly across score categories
- **Tasks:**
 - Conduct exploratory analysis of utilization distributions
 - Execute One-Way ANOVA
 - Produce summary statistics by group
 - Create visualizations showing utilization patterns
- **Finding:** Higher utilization associates with worse classification—optimal range 21-30%

Sprint Ceremonies and Continuous Feedback

Each two-week sprint included **daily stand-ups** where team members briefly reviewed:

- **Yesterday's Accomplishments** — What was completed
- **Today's Plans** — What will be attempted
- **Blockers** — What obstacles emerged

Stand-ups allowed the team to identify issues very quickly. When data quality issues that had not been anticipated through exploratory analysis, the team reacted to them instantly instead of letting them pile up. This gave way to solving the issues before they could have a negative impact on the subsequent work.

Sprint Reviews (every 2 weeks):

- Demonstrated completed work to stakeholders (instructor, peer reviewers, visualization specialists)
- Gathered feedback on preliminary findings
- Refined priorities for subsequent sprints

For the payment delay analysis sprint, this included:

- Presenting Kruskal-Wallis test results with statistical annotations
- Showing boxplots of delay distributions
- Explaining statistical significance

- Discussing implications for lending decisions

Stakeholder questions: "Can you show whether delays differ across all three score pairs?" prompted detailed Dunn's post-hoc test results with Bonferroni correction.

Sprint Retrospectives (every 2 weeks):

The team reflected on process effectiveness, generating insights that accumulated across the project:

- "Breaking work into small sprints enabled early detection of data quality issues; catching the income distribution's high skew early allowed correction before running models"
- "Peer review of analysis code caught calculation errors I would have missed; maintain this discipline"
- "Need clearer data documentation upfront to avoid redundant validation"
- "Should establish visualization templates earlier to ensure consistency"

Agile Benefits Realized During Project Execution

Early Issue Detection:

During the exploratory analysis, the team found that the income distribution was extremely skewed (26.252) in the Phase 3 analysis planning sprints. They quickly applied logging transformation and confirmed that the normality assumptions were met with the post-transformation skewness of 0.132. If the problem had been found in Phase 6 (Testing), it would have been necessary to make a lot of changes and the project would have been delayed considerably. The short sprint cycles of Agile made it possible to find and fix the problems in a matter of days instead of months (Highsmith, 2009).

Continuous Feedback Loop:

Following a payment delay analysis sprint, the team was instructed by stakeholder feedback to perform a sensitivity analysis with non-parametric Kruskal-Wallis in conjunction with parametric ANOVA alternatives in order to check the robustness of the results. This reaction intensified analytical rigor by confirming that suitable tests were chosen according to the nature of the data. The outcomes were consistent across the different methods, thus the confidence in the results was strengthened.

Incremental Value Delivery:

Stakeholders were able to grasp the initial results even tearly phases of the project. Following the completion of the analyses of payment delay and age demographics in Phase 3, the group was in a position to make the following assertion with a high level of confidence: "We have robust statistical evidence that both payment delays and age have a significant impact on credit scores; these variables should be included in lending models." This partial insight was quite useful, the project in its entirety had not been completed yet.

HYBRID FRAMEWORK: INTEGRATING WATERFALL AND AGILE

Instead of considering Waterfall and Agile as two completely different options, the project was working on two different levels which were in fact complementary. This two-level design made use of the advantages of each method and at the same time reduced the disadvantages.

Waterfall vs Agile Methodology Comparison

Contrasting structured and adaptive approaches

Aspect	Waterfall	Agile
Clarity & Planning	Upfront requirements well-defined, predictable timelines	Emergent requirements, adaptive planning
Documentation	Comprehensive, formal, extensive records	Minimal, focused on working product
Feedback Cycles	Delayed until late project phases	Continuous, every sprint
Flexibility	Rigid, costly to change mid-project	Highly adaptive to change
Resource Planning	Predictable resource allocation	Variable resource requirements
Risk Management	Late discovery of major issues	Early detection and mitigation

Level 1: Waterfall's Macro-Structure

The seven-phase Waterfall structure provided overall architecture and phase-gate discipline.

Entry Criteria for Each Phase:

- Phase 3 Analysis could not begin until Phase 2's data requirements had been fully specified and validated
- Phase 4 Program Design could not commence until Phase 3 had selected and documented all analytical methods
- This prevented the team from beginning analysis with incomplete data understanding

Exit Criteria Confirmed:

- Phase completion verified that each phase achieved its objectives before progression
- Phase 4 Program Design could not conclude until all analytical methods had been specified and documented
- This prevented implementation from beginning without clear design

Why Phase Gates Matter:

These phase gates stopped the project from going off without a clear strategy - which is a risk that is usually present in pure Agile types of work (Serrador & Pinto, 2015). The seven phases set up review points at each stage of the work. After every phase, the stakeholders looked at the progress and gave their green light to move on. Their thorough system was an execution of the most important strategic priorities being kept in line.

Level 2: Agile's Adaptive Cycles Within Phases

Within each Waterfall phase, Agile sprint cycles enabled rapid iteration, continuous learning, and stakeholder feedback integration.

Phase 3 (Analysis) Contained Four Distinct Sprints:

- **Sprint 1** — Exploratory data analysis and identifying candidates for statistical testing (1st 2 weeks)
- **Sprint 2** — Payment delay analysis in detail with Kruskal-Wallis test (weeks 3-4)
- **Sprint 3** — Age demographics examining Chi-Square test and life-stage patterns (weeks 5-6)
- **Sprint 4** — Credit utilization and inquiry frequency analysis (weeks 7-8)

By sprint's end, Phase 3 concluded with:

- Methods selected and documented
- Hypotheses tested with statistical rigor
- Preliminary insights documented
- Stakeholders aligned on analytical direction

This allowed Phase 4 (Program Design) to proceed with confidence that the analytical approach was sound and aligned with stakeholder expectations.

Why This Integration Works:

Waterfall's Macro-Structure Prevents Scope Drift:

Phase 3 was purely about analysis and did not involve any implementation or deployment activities. The members of the team decided not to start the development of machine learning models during Phase 3; instead, they left that task for Phase 5. Well-defined phases help in making sure that every element is of value before moving forward. Phase gates, as quality checkpoints, help in minimizing the risk of problems occurring at later stages (Turner, 2009).

Agile's Sprint Cycles Provide Tactical Flexibility:

The team ran into data quality issues that they weren't expecting when they were doing their data exploration; the short cycles of Agile allowed them to fix these issues right away without a big project interruption. Stakeholder feedback from the sprint reviews helped to clarify the next work. The use of an iterative approach helped to avoid the "rework surprise" that is typical of pure Waterfall projects, where a large amount of rework is found only at the late stages of the execution when it is most costly to make changes (Chen et al., 2012).

Documentation and Agility Balanced:

The waterfall model provided for very clear documentation of the analytical decisions, methods, and limitations, whereas the Agile model was more geared towards sharing the insights with stakeholders at the early stage and continuously. The regular sprint reviews kept the parties involved in the project being on the same level with the expectations and they also made it possible for the changes to be addressed at a very early stage, thus, last-minute risk was very much reduced (Schwaber & Sutherland, 2020).

CHALLENGES ENCOUNTERED AND LESSONS LEARNED

The project faced a number of time challenges that helped to highlight certain key principles when it comes to managing data analysis initiatives. They were not, however, singular challenges; instead, they are typical problems that occur in analytical projects, which have vague scope, methods, and requirements at the start.

Challenge 1: Scope Creep and Prioritization

As analytical work progressed, team members and stakeholders identified numerous additional analyses that could be conducted:

- "What about debt-to-income ratios?"
- "How do location and occupation affect credit scores?"
- "Can we build a fairness audit examining whether credit scores discriminate across demographic groups?"

Without disciplined scope management, the project could have expanded indefinitely.

How Team Addressed It:

- Maintained explicit research question list and declined analyses not directly addressing five core questions
- When colleagues suggested employment sector analysis, team referenced the charter: "Sector effects aren't one of our five core research questions. Should we add them to scope or document this as future work?"
- Most suggestions documented in "Future Work" section rather than dismissed entirely
- This approach maintained strategic focus while acknowledging valuable ideas for future investigation

Key Lesson: Teams must remain open to new insights without losing direction. Documentation of deferred ideas preserves them without consuming current project resources (Snowden & Boone, 2007).

Challenge 2: Statistical Assumption Violations

Statistical tests assume data characteristics that real-world data frequently violates:

- Shapiro-Wilk normality tests revealed many variables were not normally distributed
- Levene's test indicated variances differed significantly across groups
- Such violations can reduce reliability of traditional parametric tests like ANOVA

The Dilemma: Use parametric tests despite violations, or switch to nonparametric alternatives that might be less powerful?

How Team Addressed It:

- Explicitly documented assumption violations and justified analytical choices
- For ANOVA, despite violations, team proceeded because large sample size (n=100,000) provided robustness against moderate violations
- Supplemented ANOVA with non-parametric Kruskal-Wallis alternatives to check consistency
- Results remained similar across methods, building confidence in findings
- Transparent documentation of assumption violations built stakeholder confidence that analytical rigor was maintained despite imperfect data

Key Lesson: Rather than hiding assumption violations, acknowledge them explicitly. Document their impact and justify analytical choices. Large sample sizes provide robustness against minor violations (Grunert et al., 2005).

Challenge 3: Model Interpretability versus Accuracy Tradeoff

The Random Forest classifier got 72.8% of the cases right but did not offer much in the way of interpretability.

The Problem: Besides accuracy, Stakeholders were very much interested in knowing the reasons for the model's predictions of credit scores. "Which factors influenced that prediction the most?" Random Forest's combination of multiple decision trees makes it a very complex and obscure model that is hard to understand.

How Team Addressed It:

- Generated feature importance plots showing which variables most significantly influenced model predictions
- Top three features: Outstanding Debt, Changes in Credit Limit, Annual Income
- Feature importance scores (mean decrease in impurity across all trees) provided insight into variable relevance
- Documentation discussed the interpretability-accuracy tradeoff: accuracy gains from complex ensemble methods come at the cost of interpretability

Key Lesson: Perfect transparency isn't always achievable. Communicate tradeoffs clearly and use interpretability techniques (feature importance, SHAP values) to provide insight into black-box models (Hand & Henley, 1997).

Challenge 4: Documentation Burden and Time Competition

Comprehensive documentation of analytical procedures, statistical tests, and interpretation rationales requires substantial time and discipline. During sprints focused on completing analyses, the temptation to shortchange documentation proved strong: "We're running out of time; we can document this later."

Why This Matters: Projects without adequate documentation face "knowledge debt" that limits future replication, extension, and learning.

How Team Addressed It:

- Treated documentation as a first-class deliverable, reserving 15-20% of sprint effort specifically for it
- Code thoroughly commented explaining transformations
- Analysis notebooks combined code, outputs, and narrative explanations
- README files documented procedures and data
- This discipline ensured that when sprints concluded, analyses could be understood by future team members or stakeholders seeking to extend the work

Key Lesson: Documentation investment pays dividends. When future questions arise about analytical decisions, comprehensive records enable quick answers without re-analysis (Kerzner, 2017).

Lessons Learned

1. **Hybrid Integration Works Best:** Waterfall provided structure and clear phase boundaries, while Agile supported flexibility during execution. Together, they complemented each other and improved overall project control (Serrador & Pinto, 2015).

2. **Small Units of Work Reduce Risk:** Short, clearly defined work cycles improved progress visibility, enabled faster issue detection, and allowed continuous feedback, preventing problems from accumulating.
3. **Continuous Stakeholder Engagement Matters:** Regular reviews kept stakeholders aligned with evolving findings and reduced the risk of late-stage misunderstandings (Highsmith, 2009).
4. **Data Quality Comes First:** Early validation of data prevented flawed analysis and avoided costly corrections after models were already built (Arya et al., 2013).
5. **Documentation Adds Long-Term Value:** Recording analytical decisions and limitations improved transparency and made future extensions of the work more feasible (Turner, 2009).

RECOMMENDATIONS FOR PRACTICE

Data Analysis projects are best handled through a hybrid methodology that integrates Waterfall phase structure with Agile execution. The well-defined phase gates assist in scope control whereas the short sprints within the phases enable the teams to address quickly the data issues and the stakeholder feedback (Conforto & Amaral, 2016).

Breaking down work into small, testable units greatly enhances the visibility of progress, speeds up the learning process, and lowers the risk that has been accumulated. Involving stakeholders on a regular basis through reviews helps in expectation management and thus, expensive revisions in the final stages of the project are avoided (Schwaber & Sutherland, 2020; Cohn, 2010).

It is better for the team to think of documentation and assumption validation as their main tasks rather than just overhead. Having clear records of methods, assumptions, and limitations not only makes the project more transparent, but it also helps in reproducing the work, and it stays valuable for a long time (Turner, 2009; Snowden & Boone, 2007).

Lastly, the selection of tools should be done with a great consideration of how they will facilitate collaboration, tracking, and reproducibility. Training should be provided to ensure that the tools are work facilitators and not that they add to the complexity (Kerzner, 2017).

CONCLUSION

Data analysis projects are less certain and that uncertainty is not well handled by either Waterfall or Agile if taken separately. Waterfall brings the necessary structure as it outlines the scope, methods, and quality gates in advance, whereas Agile helps to maintain the flexibility by having short feedback cycles and continuous stakeholder engagement (Beck et al., 2001). In fact, the two are not opposing forces but rather different levels where the two approaches work when merged.

This project showed how a hybrid model can be of real practical use:

- The Waterfall phases gave an overall framework to the work, which was guided from the requirements to the final delivery and the drift of the scope was prevented.
- The definite goals of the phase and exit criteria guaranteed that the quality standards were recognized before the progression.
- Agile sprints performed in each phase allowed rapid iteration, early issue detection, and incremental insight delivery.

- The engagement of stakeholders on a regular basis ensured the alignment and thus the rework at the late stage was minimized (Serrador & Pinto, 2015).

Organizations should not decide Waterfall or Agile, but rather use hybrid frameworks which combine the advantages of both. Some of the features of a successful implementation are:

- Well structured Waterfall stages with their own criteria for entry and exit.
- Agile sprints integrated with frequent feedback.
- Small, testable units of work.
- Continuous stakeholder involvement.
- Robust documentation and assumption validation activities.

By combining Waterfall's structure with Agile's flexibility, organizations can manage analytical uncertainty while still delivering reliable, goal-driven outcomes (Conforto & Amaral, 2016).

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